". . . Bart! Hello, Bart!" Lisa yells through the wormhole. "Have you been listening to me? Step on it. I want to get home in time for dinner." Bart looks into his wormhole mouth and tells Lisa he's already landed on the lawn of Springfield High. Looking more closely through the wormhole, Lisa sees that Bart is telling the truth, but looking out of the Kwik-E-MART toward Springfield High, she doesn't see his ship on the lawn. "I don't get it," she says.

"Actually, it makes perfect sense," Bart proudly answers. "I've landed at Springfield High, but 6 million years into the future. You can't see me by looking out the Kwik-E-Mart window, because you're looking at the right place, but you're not looking at the right time. You're looking 6 million years too early."

"Oh, right, that time-dilation thing of special relativity," Lisa agrees. "Cool. Anyway, I want to get home in time for dinner, so climb through the wormhole, because we've got to hurry." "Okay," Bart says, crawling through the wormhole. He buys a Butterfinger from Apu, and he and Lisa head home.

Notice that although Bart's passage through the wormhole took him but a moment, *it transported him 6 million years back in time.* He and his ship and the wormhole mouth had landed far into earth's future. Had he gotten out, spoken with people, and checked the newspaper, everything would have confirmed this. Yet, when he passed through the wormhole and rejoined Lisa, he found himself back in the present. The same holds true for anyone else who might follow Bart through the wormhole mouth: he would also travel 6 million years back in time. Similarly, anyone who climbs into the wormhole mouth at the Kwik-E-Mart, and out of the mouth Bart left in his ship, would travel 6 million years into the future. The important point is that Bart did not just take one of the wormhole mouths on a journey through space. His journey also transported the wormhole mouth through time. *Bart's voyage took him and the wormhole'smouth into earth's future. In short, Bart transformed a tunnel through space into a tunnel through time; he turned a wormhole into a time machine.*

A rough way to visualize what's going on is depicted in Figure 15.5. In Figure 15.5a we see a wormhole connecting one spatial location with